

Bagging and Boosting (Ensemble Learning)

Ensemble Learning

Ensemble learning in Machine Learning is a technique where multiple models (weak learners) are combined to form a stronger and more accurate model. It helps improve prediction performance and robustness.

1. Bagging (Bootstrap Aggregating)

Definition

Bagging is an ensemble technique in which multiple models are trained independently using different subsets of the training data generated through bootstrap sampling, and their outputs are aggregated.

Bootstrap Sampling

- Random sampling with replacement
- Each subset has the same size as the original dataset
- Some data points may repeat, while others may be left out

Working Mechanism

1. Generate multiple bootstrap samples from the original dataset
2. Train a separate model on each sample
3. Combine predictions:
 - Classification → Majority Voting
 - Regression → Averaging

Mathematical Representation

If there are n models:

- Regression:
Final Prediction = $(f_1(x) + f_2(x) + \dots + f_n(x)) / n$
- Classification:
Final Prediction = $\text{Mode}(f_1(x), f_2(x), \dots, f_n(x))$

Characteristics

- Models are **independent**
 - Training can be done **in parallel**
 - Reduces model instability
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Boosting

Definition

Boosting is an ensemble technique where models are trained sequentially, and each new model focuses on correcting the errors made by the previous models.

Core Idea

- Assign higher importance (weights) to misclassified data points
- Each new model learns from the mistakes of previous models

Working Mechanism

1. Initialize equal weights for all data points
2. Train the first model
3. Identify incorrectly predicted samples
4. Increase their weights
5. Train the next model on updated weights
6. Repeat the process
7. Combine all models into a final prediction

Mathematical Representation

Final model is a weighted combination of weak learners:

$$F(x) = w_1 f_1(x) + w_2 f_2(x) + \dots + w_n f_n(x)$$

- - Focus on error correction
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Types of Boosting

Gradient Boosting

- Optimizes loss function using gradient descent
- Learns from residual errors

XGBoost

- Improved version of gradient boosting
 - Includes regularization and faster computation
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